Pharmaceutical Solutions

- Types of Solutions
- Concentration Expression
- Ideal and Real Solutions
- Colligative Properties
- Application of Solutions in Physical Pharmacy

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PHARMACEUTICAL SOLUTIONS

- Pharmaceutical solutions are homogeneous liquid dosage forms in which the drug is completely dissolved in a suitable solvent or solvent system.
- **Physical pharmacy relevance**
  Behavior of solutions is governed by solubility, dissolution, molecular interaction, and stability principles.
- **Single-phase systems**
  Unlike suspensions or emulsions, solutions contain only one phase with uniform drug distribution.
- **Importance in dosage form science**
  Solutions provide rapid onset of action and uniform dosing when properly formulated.
- **Routes of administration**
  Can be administered oral, parenteral, ophthalmic, otica, or topical.
- **Clinical significance**
  Especially useful when fast therapeutic response or dose flexibility is required.

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Characteristics of Pharmaceutical Solutions

- **Homogeneity**
  Drug molecules are evenly distributed at molecular level throughout the solvent.
- **Clarity**
  Solutions are usually clear and transparent unless colored by drug or excipients.
- **Uniform drug content**
  Each unit volume contains identical drug concentration, ensuring dose accuracy.
- **Rapid absorption potential**
  Drug is already in dissolved state, eliminating dissolution step before absorption.
- **Ease of administration**
  Suitable for patients with swallowing difficulties.
- **Formulation sensitivity**
  Sensitive to pH, temperature, light, and microbial contamination.

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Role of Solutions in Physical Pharmacy

- **Solubility principles**
  Drug solubility determines feasibility of solution dosage form.
- **Molecular interactions**
  Drug–solvent interactions influence stability and solubilization.
- **pH dependency**
  pH affects drug ionization and solution stability (details covered later lectures).
- **Thermodynamic stability**
  Solutions are thermodynamically stable systems if drug remains dissolved.
- **Diffusion behavior**
  Drug molecules diffuse freely, aiding rapid absorption.
- **Foundation for advanced systems**
  Many parenteral and ophthalmic products are solution-based.

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Advantages of Pharmaceutical Solutions

- **Rapid onset of action**
  No disintegration or dissolution step required after administration.
- **Dose flexibility**
  Dose can be adjusted by measuring volume, useful in pediatrics.
- **Uniform dosing**
  Homogeneity ensures consistent drug delivery.
- **Ease of swallowing**
  Suitable for pediatric, geriatric, and unconscious patients (oral/enteral).
- **Versatility of routes**
  Can be formulated for multiple administration routes.
- **Reduced local irritation (when diluted)**
  Proper formulation can reduce drug irritation.

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Disadvantages of Pharmaceutical Solutions

- **Chemical instability**
  Drugs degrade faster in solution due to hydrolysis or oxidation.
- **Microbial growth risk**
  Aqueous solutions require preservatives.
- **Bulky packaging**
  Liquids require larger containers than solids.
- **Taste issues**
  Bitter drugs require sweeteners or flavors.
- **Shorter shelf life**
  Compared to solid dosage forms.
- **Transportation issues**
  Risk of leakage and spillage.

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Classification of Pharmaceutical Solutions

- **Based on route of administration**
  Oral, parenteral, ophthalmic, otic, topical solutions.
- **Based on solvent system**
  Aqueous and non-aqueous solutions.
- **Based on therapeutic use**
  Systemic and local action solutions.
- **Based on formulation purpose**
  Ready-to-use and reconstituted solutions.
- **Based on special requirements**
  Sterile and non-sterile solutions
- **Based on solute concentration**
  Saturated, unsaturated and supersaturated solutions

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Classification Based on Route of Administration

- **Oral solutions**
  Administered via mouth for systemic or local GI action.
- **Parenteral solutions**
  Injected directly into body fluids; must be sterile and pyrogen-free.
- **Ophthalmic solutions**
  Applied to eyes; require sterility and isotonicity.
- **Otic solutions**
  Used in ear canal for local action.
- **Topical solutions**
  Applied to skin or mucous membranes.
- **Special cavity solutions**
  Nasal and throat solutions.

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Oral Pharmaceutical Solutions

Liquid solutions intended for oral administration where drug is completely dissolved.

- **Purpose**
  Provide systemic or local gastrointestinal effect.
- **Patient suitability**
  Ideal for pediatric and geriatric patients.
- **Taste considerations**
  Require sweeteners and flavors for patient acceptance.
- **Stability considerations**
  pH and preservatives are critical.
- **Example (API)**
  Paracetamol oral solution.

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Parenteral Solutions

Sterile solutions intended for injection into the body.

- **Sterility requirement**
  Must be free from microorganisms and pyrogens.
- **Rapid therapeutic action**
  Direct entry into systemic circulation.
- **Physiological compatibility**
  pH and tonicity must be controlled.
- **Packaging requirements**
  Ampoules or vials with airtight sealing.
- **Example (API)**
  Dextrose injection.

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Ophthalmic Solutions

Sterile solutions intended for application to the eye.

- **Sensitivity of site**
  Eye tissues are highly sensitive to irritation.
- **Sterility importance**
  Prevents eye infections.
- **Clarity requirement**
  Must be free from particles.
- **pH considerations**
  Adjusted close to physiological pH.
- **Example (API)**
  Ciprofloxacin eye drops.

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Otic Solutions

- Used for local antimicrobial, anti-inflammatory, or analgesic action.
- **Sterility status**
  Usually non-sterile but must be clean and safe.
- **Viscosity requirement**
  Slight viscosity improves retention.
- **Local action**
  Minimal systemic absorption intended.
- **Example (API)**
  Chloramphenicol ear drops.

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Topical Solutions

- Solutions applied to skin for local therapeutic effect.
- **Rapid drying**
  Volatile solvents evaporate quickly after application.
- **Ease of spreading**
  Cover large skin surface easily.
- **Limited penetration**
  Primarily act on surface layers.
- **Irritation potential**
  Alcoholic solutions may irritate damaged skin.
- **Example (API)**
  Povidone iodine solution.

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Classification Based on Solvent System

- **Aqueous solutions**
  Use water as primary solvent.
- **Non-aqueous solutions**
  Use solvents other than water.

Solvent selection importance
Affects solubility, stability, and safety.

- **Toxicity considerations**
  Solvent must be safe for intended route.
- **Regulatory acceptance**
  Only approved solvents can be used.

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Aqueous Pharmaceutical Solutions

Solutions in which water is the main solvent.

- **Advantages**
  Non-toxic, economical, and widely accepted.
- **Stability concern**
  Susceptible to hydrolysis and microbial growth.
- **Preservative requirement**
  Essential for multi-dose preparations.
- **Common use**
  Oral, ophthalmic, parenteral solutions.
- **Example**
  Oral Rehydration Solution.

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Non-Aqueous Pharmaceutical Solutions

- Solutions prepared using solvents other than water.
- **Purpose**
  Used when drug is unstable or poorly soluble in water.
- **Improved stability**
  Reduce hydrolytic degradation.
- **Viscosity effect**
  Higher viscosity may slow drug release.
- **Safety concern**
  Solvent toxicity must be evaluated.
- **Example**
  Alcoholic iodine solution.

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Classification Based on Therapeutic Use

- **Systemic solutions**
  Drug absorbed into circulation to produce body-wide effect.
- **Absorption requirement**
  Rapid absorption desired.
- **Local action solutions**
  Intended to act at site of application.
- **Minimal systemic absorption**
  Reduces systemic side effects.
- **Formulation difference**
  Local solutions focus on retention and safety.
- **Examples**
  Systemic: Oral paracetamol solution
  Local: Antiseptic throat solution.

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Classification based on formulation

Reconstituted Solutions

Solutions prepared immediately before use by dissolving powder in solvent.

- **Reason for use**
  Improves stability of drugs unstable in solution.
- **Storage advantage**
  Dry form has longer shelf life.
- **User responsibility**
  Requires proper reconstitution technique.
- **Common use**
  Antibiotics and pediatric formulations.
- **Example**
  Dry amoxicillin powder for reconstitution.

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Types Of Solutions Based On Degree Of Saturation

- **Unsaturated Solution**

An unsaturated solution contains less solute than the maximum amount that can dissolve in the solvent at a given temperature. More solute can still dissolve without any precipitation.

- Solute concentration < solubility limit
- Stable system
- Most pharmaceutical solutions are prepared in this form
- **Example:**
  Sugar in water where more sugar can still dissolve

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Saturated Solution

- A saturated solution contains the maximum amount of solute that can dissolve in the solvent at a specific temperature. Any additional solute added will remain undissolved.
- Solute concentration = solubility limit
- Exists in equilibrium with undissolved solute
- Solubility depends on temperature
- **Example:**
  Saturated sodium chloride solution at room temperature

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Supersaturated Solution

A supersaturated solution contains more solute than the normal at a given temperature.
It is thermodynamically unstable.

- Prepared by heating a saturated solution and cooling slowly
- Excess solute may crystallize on shaking or seeding
- Metastable system
- **Example:**
  Supersaturated sugar syrup

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Stability Issues in Pharmaceutical Solutions

- **Chemical degradation**
  Hydrolysis and oxidation reduce drug potency.
- **Photodegradation**
  Light-sensitive drugs degrade in clear containers.
- **Microbial contamination**
  Aqueous solutions support microbial growth.
- **Temperature effects**
  High temperature accelerates degradation.
- **pH sensitivity**
  pH changes affect stability.
- **Packaging**
  Amber bottles and airtight containers improve stability.

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Concentration Expression

Concentration is the amount of drug (solute) present in a given quantity of solution or solvent.
Why concentration matters in pharmacy

- It decides dose accuracy
- It determines drug effectiveness
- It affects safety of the patient
- Example
- If a syrup contains 250 mg drug in 5 mL, the patient receives 250 mg every 5 mL dose.
- If concentration is wrong , patient gets overdose or under dose.

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Amount of Drug

- **Amount of drug = quantity of drug substance present**
- It can be expressed as:

Expression	Used in
Mass (mg, g)	Clinical dosing
Volume (mL)	Liquids
Moles	Physical pharmacy
Equivalents	Electrolytes

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Why Physical Pharmacy Uses Molecular Amount

Drug behavior depends on number of molecules, not just weight

- **Example**
- 500 mg Paracetamol
- 500 mg Insulin
- ❌ Same weight
  Different number of molecules
  Different biological behavior

Why?

- Because their **molecular weights are different**

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MOLARITY (M)

Molarity is defined as

The number of **moles of solute present in one liter (1000 mL) of solution**

Molarity (M)=Number of moles of solute/Volume of solute in liters

•Total final volume of solution is considered
•Volume includes solute + solvent
•Solute amount is expressed in moles

Characteristics of Molarity
•Volume-based
•Temperature dependent
•Changes with expansion or contraction of solution
•Commonly used in chemical reactions

Pharmaceutical Importance
•Preparation of buffer solutions
•Analytical chemistry
•Assay calculations

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Example

- Prepare **1 M NaCl solution**:
- Molecular weight of NaCl = 58.5 g/mol
- Weigh **58.5 g NaCl**
- Dissolve and **make volume up to 1 liter**

Important

Volume of solution, not solvent
Temperature affects molarity
Used mainly in chemistry-based calculations

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Introduction to the Mole

- **What is a mole?**
- A **mole** is a unit that represents a **fixed number of particles**.
- **Definition**
- **One mole** of a substance contains **6.022×10^{23} particles**
- These particles can be:
- Molecules
- Atoms
- Ions

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MOLALITY (m)

Molality is defined as

The number of **moles of solute present in one kilogram (1000 g) of solvent**

Molality (m)=Number of moles of solute/Weight of solvent in Kg

Solute amount is in moles
Solvent amount is in mass, not volume
Only solvent is considered, not total solution

Characteristics of Molality
Mass-based
Temperature independent
More accurate for theoretical calculations

Pharmaceutical Importance
Used in colligative properties
Essential for tonicity calculations
Stability studies
Physical pharmacy principles

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Example

- Prepare 1 m glucose solution:
- Molecular weight of glucose = 180 g/mol
- Weigh 180 g glucose
- Dissolve in 1 kg of water (not 1 L)
- Final volume is irrelevant

- Uses **kg of solvent**, not solution
  ✔ Independent of temperature

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NORMALITY (N)

Normality is defined as
The number of **gram equivalents of solute present per liter of solution**

Normality (N)=Gram equivalents/Volume of solution in liters

Equivalent weight=Molecular weight/Valance

Normality depends on:
Nature of solute
Type of reaction (acid–base, redox, precipitation)
Same solution can have different normality in different reactions

Pharmaceutical Importance
Acid–base titrations
Assay of drugs
Analytical procedures

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Example

- Prepare 1 N HCl:
- Molecular weight of HCl = 36.5
- Valence = 1
- Equivalent weight = 36.5

→ Dissolve 36.5 g HCl and make volume to 1 liter

- Reaction-specific
  Used mainly in **analysis**
  Less common in formulations

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PERCENTAGE CONCENTRATION EXPRESSIONS

- Percentage Weight in Weight (% w/w)

  %w/w= Weight of solute/Weight of solution×100

Grams of solute in **100 g of solution**

Uses in Pharmacy

- Ointments
- Creams
- Powders
- Solid preparations
- **Example**
- 5% w/w ointment

5 g drug in 100 g ointment

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Percentage Weight in Volume (% w/v)

%w/v=Weight of solute/Volume of Solution×100
Grams of solute in **100 mL of solution**

Uses in Pharmacy

- Syrups
- Oral liquids
- Injections
- IV fluids

Example

- 5% w/v glucose
  5 g glucose in 100 mL solution

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Percentage Volume in Volume (% v/v)

- %v/v=Volume of solute/Volume of solution×100

Final volume must be **100 mL**
Do not add volumes directly

Uses in Pharmacy

- Alcoholic solutions
- Elixirs
- Spirits

Example

70% v/v alcohol
Measure 70 mL alcohol
Add solvent **q.s. to make 100 mL**

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PARTS PER MILLION (ppm)

- Number of parts of solute per **1,000,000 parts of solution**

Practical Pharmaceutical Meaning

- For dilute aqueous solutions

1 ppm≈ 1mg/L

Uses in Pharmacy

- Trace impurities
- Preservative limits
- Water quality
- Toxicological studies

Limitations

Valid only for very dilute solutions

Density must be approximately 1 g/mL

Example

- 2 ppm fluoride
  → 2 mg fluoride per liter of water

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NUMERICALS ON MOLARITY (M)

- **General Formula for Molarity**
- Molarity (M)=Number of moles of solute/Volume of solution in li ters× 100
- Rearranging for weight:

Weight of solute (g)=M× Molecular weight×Volume (L)

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Lets solve

- **Example 1: Preparation of 1 M NaCl Solution (Standard)**
- **Given:**
- Required molarity (M) = 1 M
- Molecular weight of NaCl = 58.5 g/mol
- Volume of solution = 1 L
- Formula used:

$$\text{Weight} = M \times \text{Molecular weight} \times \text{Volume}$$

Calculation:

- Weight=1×58.5×1 Weight=58.5 g
- Answer:
  Weigh 58.5 g of NaCl, dissolve in water, and **make the volume up to 1 liter.**

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Example 2

Preparation of 0.5 M NaCl Solution

- **Given:**

Required molarity (M) = 0.5 M
Molecular weight of NaCl = 58.5 g/mol
Volume of solution = 1 L

- Formula used:

  Weight=M× Molecular weight× Volume

- Calculation:

  Weight=0.5×58.5×1

- Weight=29.25 g Answer:
  Weigh 29.25 g of NaCl, dissolve in water, and make the volume up to 1 liter.

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Example 3

- Preparation of **0.5 M NaCl Solution (500 mL)**

Given:

- Required molarity = 0.5 M
- Molecular weight of NaCl = 58.5 g/mol
- Volume = 500 mL = **0.5 L**
- Formula used:

  Weight=M× Molecular weight×Volume

Calculation:

  Weight=0.5×58.5×0.5

- Weight=14.625 g Answer:
  Weigh 14.625 g of NaCl, dissolve in water, and make the volume up to 500 mL.

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Example 4

Preparation of **0.1 M NaCl Solution (250 mL)**

Given:

- M = 0.1
- Molecular weight = 58.5 g/mol
- Volume = 250 mL = 0.25 L

Formula used:

- Weight=M× Molecular weight×Volume

Calculation:

- Weight=0.1×58.5×0.25
- Weight=1.4625 g

Answer:

Weigh 1.46 g of NaCl, dissolve and make volume up to 250 mL.

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Important

✔ Always write the formula first
✔ Convert mL to liters
✔ Use final solution volume
✔ Units must be shown
✔ Round off only at the end

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MOLALITY (m) NUMERICALS

- Formula:
  Molality (m) = Number of moles of solute / Weight of solvent (kg)

  Rearranged:
  Weight of solute (g) = m × Molecular weight × Weight of solvent (kg)

- Example 1: Prepare 1 m NaCl solution using 1 kg water
  Given:
  m = 1
  Molecular weight of NaCl = 58.5 g/mol
  Weight of solvent = 1 kg

  Calculation:
  Weight = 1 × 58.5 × 1
  Weight = 58.5 g

  Answer:
  Weigh 58.5 g NaCl and dissolve in 1 kg water.

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Examples

- Example 2: Prepare 0.5 m NaCl solution using 1 kg water
  Calculation:
  Weight = 0.5 × 58.5 × 1
  Weight = 29.25 g

- Example 3: Prepare 0.5 m NaCl solution using 500 g water
  Calculation:
  Weight = 0.5 × 58.5 × 0.5
  Weight = 14.625 g

- Example 4: Prepare 1 m glucose solution using 1 kg water
  Molecular weight of glucose = 180 g/mol

  Calculation:
  Weight = 1 × 180 × 1
  Weight = 180 g

- Example 5: Prepare 0.2 m glucose solution using 250 g water
  Calculation:
  Weight = 0.2 × 180 × 0.25
  Weight = 9 g

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NORMALITY (N) NUMERICALS

- Formula:
  Normality (N) = Equivalents / Volume (L)

  Equivalent weight = Molecular weight / n-factor

  Rearranged:
  Weight (g) = N × Equivalent weight × Volume (L)

- Example 1: Prepare 1 N HCl (1 L)
  Molecular weight = 36.5
  n-factor = 1
  Equivalent weight = 36.5

  Calculation:
  Weight = 1 × 36.5 × 1
  Weight = 36.5 g

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Examples

- Example 2: Prepare 0.5 N HCl (500 mL)
  Calculation:
  Weight = 0.5 × 36.5 × 0.5
  Weight = 9.125 g

- Example 3: Prepare 1 N H2SO4 (1 L)
  Molecular weight = 98
  n-factor = 2
  Equivalent weight = 49

  Calculation:
  Weight = 1 × 49 × 1
  Weight = 49 g

- Example 4: Prepare 0.2 N H2SO4 (250 mL)
  Calculation:
  Weight = 0.2 × 49 × 0.25
  Weight = 2.45 g

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Examples

- Example 6: Prepare 0.5 N $Ca(OH)_2$ (1 L)
  Molecular weight = 74
  n-factor = 2
  Equivalent weight = 37

  Calculation:
  Weight = 0.5 × 37 × 1
  Weight = 18.5 g

- Example 7: Calculate normality if 4.9 g H_2SO_4 is present in 250 mL
  Equivalents = 4.9 / 49 = 0.1
  Normality = 0.1 / 0.25 = 0.4 N

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PERCENTAGE NUMERICALS

- Percentage w/w:
  % w/w = (Weight of solute / Weight of solution) × 100

  Example 1: Prepare 5% w/w ointment (200 g)
  Solute = (5 × 200) / 100 = 10 g
  Base = 190 g

- Example 2: Calculate % w/w if 12 g drug is in 240 g mixture
  % w/w = (12 / 240) × 100 = 5%

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Examples

- Percentage w/v:
  % w/v = (Weight of solute / Volume of solution) × 100

  Example 3: Prepare 5% w/v glucose solution (500 mL)
  Solute = (5 × 500) / 100 = 25 g

- Example 4: Prepare 0.9% w/v NaCl (1000 mL)
  Solute = (0.9 × 1000) / 100 = 9 g

- Example 5: Calculate % w/v if 2.5 g drug is in 50 mL
  % w/v = (2.5 / 50) × 100 = 5%

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Examples

- Percentage v/v:
  % v/v = (Volume of solute / Volume of solution) × 100

  Example 6: Prepare 70% v/v alcohol (100 mL)
  Alcohol = 70 mL, add water q.s. to make 100 mL

- Example 7: Prepare 20% v/v glycerin (250 mL)
  Glycerin = (20 × 250) / 100 = 50 mL

- Example 8: Calculate % v/v if 30 mL alcohol is in 150 mL solution
  % v/v = (30 / 150) × 100 = 20%

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PARTS PER MILLION (ppm) NUMERICALS

- Formula:
  ppm = mg of solute / liter of solution (for dilute aqueous solutions)

- Example 1: Calculate ppm if 2 mg drug is in 1 L water
  ppm = 2 / 1 = 2 ppm

- Example 2: Calculate ppm if 5 mg fluoride is in 2 L water
  ppm = 5 / 2 = 2.5 ppm

- Example 3: Prepare 10 ppm solution (1 L)
  Drug required = 10 mg

- Example 4: Prepare 25 ppm solution (500 mL)
  Drug required = 25 × 0.5 = 12.5 mg

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Examples

- Example 5: Calculate ppm if 0.02 g drug is in 4 L water
  0.02 g = 20 mg
  ppm = 20 / 4 = 5 ppm

- Example 6: Calculate ppm if 3 mg solute is in 3 kg water
  ppm = (3 / 3,000,000) × 10^6 = 1 ppm

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